APPROVED BRITISH UNIVERSITY VIETNAM BUV

18 DEC 2024

BRITISH UNIVERSITY VIETNAM BUV

In collaboration with Staffordshire University

Module Title:	Software Development
Module Code:	COMP40053
Attempt:	First Sit
Group or Individual:	Individual Assignment
Assignment Number:	1 of 3
Assignment valid until:	February 2026
Assignment Title:	Practical Assessment
Weighting:	60%
Assignment set by:	Fraser Harrison
Assignment verified by:	Donie Jardeleza

Date Issued to Students:	TBA
Submission Date and Time:	See Canvas module homepage for submission date
Method of Submission:	Portfolio Presentation

Learning Outcomes to be assessed:

1. Design and develop solutions to problems using appropriate programming paradigms and programming languages.

   Application, Problem Solving, Analysis

2. Design and implement an application with an appropriate interface and interactions using current web standards and practices

   Problem Solving, Application

3. Plan a test-and-deploy strategy, and apply it to an application to assess its quality.

   Learning, Application, Enquiry

AI Level:

The table below shows to what extent you are allowed to use Generative AI (GenAI) tools such as ChatGPT in your assessment. Please refer to the BUV Student AI Guidelines for full information how you can use GenAI tools in an ethical manner at BUV.

AI Level	Description	This assessment
1. No AI	The assessment is completed entirely without AI assistance. This level ensures that students rely solely on their knowledge, understanding, and skills. AI must not be used at any point during the assessment.	
2. AI Assisted idea generation and structuring	AI can be used in the assessment for brainstorming, creating structures, and generating ideas for improving work. No AI content is allowed in the final submission.	
3. AI Assisted language editing	AI can be used to make improvements to the language of student created work to improve the quality of the final output, but no new content can be created using AI. AI can be used, but your original work with no AI content must be provided in an appendix.	
4. AI task completion, human evaluation	AI is used to complete certain elements of the task, with students providing discussion or commentary on the AI-generated content. This level requires critical engagement with AI generated content and evaluating its output. You will use AI to complete specified tasks in your assessment. Any AI created content must be cited. You must not use AI to support with your commentary or evaluation of the work.	X
5. Full AI	AI should be used as a 'co-pilot' in order to meet the requirements of the assessment, allowing for a collaborative approach with AI and enhancing creativity. You may use AI throughout your assessment to support your own work and do not have to specify which content is AI generated.	

Generative AI instruction

AI should be used to support your development. In both structure and debugging. The design and code should be of your own implementation and concept.

Assessment Brief

Task:

You must develop a Command Line Interface Java application built with Object Orientated Programming (OOP) principles. Your code will take simple text entry from the user and fetch information from a database before doing some calculations and returning formatted information to the user. You will demonstrate your program to the teaching team and answer questions on the code submitted.

Scenario:

You are tasked with developing a software application to assist university administration in calculating the final grades for students at university. Manually calculating final grades can be time-consuming and prone to errors. To streamline this process and ensure accuracy, you will develop a program using object-oriented programming (OOP) principles. The tool should provide functionality to calculate assignments of different types for different types of students and calculate final grades based on a weighted average.

This assignment is based upon the academic regulations of Staffordshire University. It does not however include processes such as resits, exceptional circumstances, appeals and raising classifications. It does not adequately meet the requirements for classifying your degree and should be used for informational purposes only.

Please note: Appendix 3 contains a list of examples of grade calculations you may use to double check your understanding of the rules.

Components of a degree

Degrees are made up of 18 modules

- 6 modules at Level 4 (Year 1)
- 6 modules at Level 5 (Year 2)
- 6 modules at Level 6 (Year 3)

Individual Module and Assignments

Each module will receive a mark out of 100. These grades will be based on the weighted average of 1 or more assessments.

For example, Introduction to Business is made up of three assessments

- A presentation weighted at 30%
- A written report weighted at 10%
- A final exam weighted at 60%

Scores for individual Assignments

Each type of assessment is scored differently.

- Presentations and reports are scored on a rubric out of 100
- Exams have varying amount of total marks available to them and both the marks scored by the student and the total available marks are stored in the dataset

Passing a Module

A student does not need to pass every assignment to pass a module. However they must attempt every assignment. **If they do not attempt an assignment and do not submit, they will not pass the module even if the total average is above 40%.** This should be considered an overall score of 0 for the assignment

Final degree classification

Final degree classification is based on only level 5 and level 6 modules. All the grades in level 5 and 6 are looked at and the lowest is removed.

Once your lowest mark is removed, the average (Mean) mark at each level is calculated. Finally, these levels are averaged using a weighting. Level 6 are weighted at 70% and level 5 are weighted at 30%. This final average is then scored against the grade classification table

Student Grade	Grade Classification
0 – 39	Fail
40 – 49	3rd Class Pass
50 - 59	2nd class 2nd Division
60-69	2nd Class 1st Division
70+	First Class

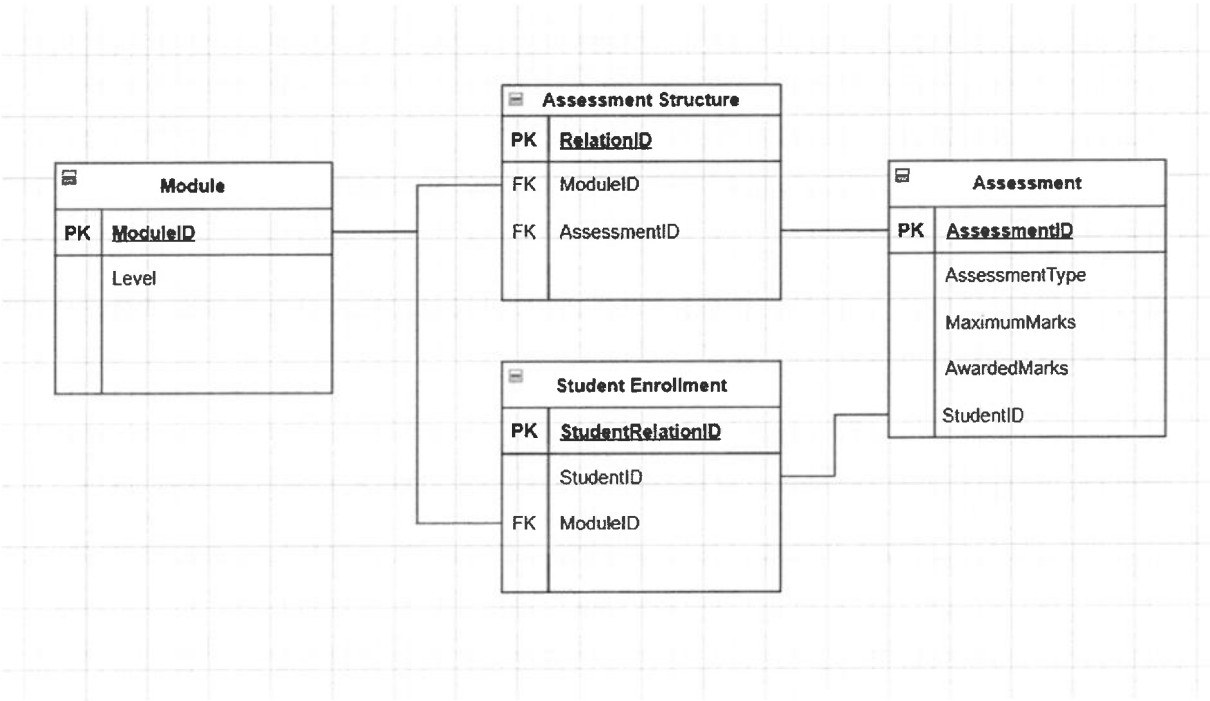
Below are a list of functions that should be completed

Task	Explanation	Complexity	Conditions tested
Get a student grade for a specific assignment	Write a function that accepts a student ID, Module ID and Assignment ID and returns the student grade	Simple	• Valid request returns a grade for a presentation • Valid request for an exam returns a student mark, the total marks available and a grade out of 100 • Invalid request is not accepted

Get a weighted average grade for a module for a specific module	Write a function that accepts a student ID, Module ID and returns the module grade based on the weighted average	Intermediate	• Valid request for a module with single 100% assignment returns correct grade • Valid request for a module with two 50% assignments return the correct grade • Valid request for a module with multiple assignments of different weightings returns the correct grade
Get a final degree classification	Write a function that accepts a student ID and returns the final degree classification.	Advanced	• Valid request will calculate final degree classification

Dataset

There is an available SQLite database to help you with this assignment. The assignment is marked on your code implementation and so the database is deliberately simplified



The Module table contains a list of modules and their levels either 4, 5 or 6

The Assessment structure table contains a list of which assessments belong to each Module and the weighting of each Assessment

The Student Enrolment table contains a list of Students enrolled onto modules

The Assessment table contains

- AssessmentType – Report, Presentation or Exam
- MaximumMarks – This will only have a value for exams. Otherwise it will be null
- AwardedMarks – The amount of marks awarded to the student
- StudentID – A reference to the student who sat the Assessment

Guidance Notes

a) Do not include your name anywhere on your assignment.

b) When you upload your file for submission you must name the file by your full student number in the following format: 'HANXXXXXXXXX'

c) Your source code should be submitted as a zipped file containing all of your code and may include a readme txt with any references

d) The marking scheme that will be used to mark your work is shown at the end of this document in Appendix 1.

Submission Procedure

You are not required to print and submit a hard copy of your work, as all submissions are carried out electronically through Canvas. Please ensure you do not wait until the last minute to carry out your submission; computer problems can happen any time and this will not be accepted as an excuse for late submission. This includes internet connectivity issues. When you submit your assignment, it is in your best interests to save and/or print a copy of your online submissions receipt to prove that you have submitted your assignment on time.

If you feel that you can't submit your assessment on time, you should speak to your module leader who will help you to work out how you can hand-in on time.

If unexpected circumstances which you could not have planned for have happened, you may be able to claim for exceptional circumstances to be taken into account. There is more information here.

If you have no exceptional circumstances and you do not submit you will fail the assessment. If you submit within 5 working days of the hand-in deadline we will mark your work as late, meaning the maximum mark you can achieve is the basic pass mark and this will count as your first attempt. If you are taking a re-sit assessment, late submissions will not be accepted.

Please see the Staffordshire university regulations here for more information

The Maximum word limit is shown on the front cover of the assignment.

You must provide an accurate word count at the end of your assignment. This word count must be placed before your reference list and should include all the written content of your assignment **excluding** the words used in any cover pages, contents pages, titles, sub-titles, reference list and appendices.

In the assignment, in addition to the word limit, you may make sensible use of tables, images of academic models, diagrams, figures and student-created graphics such as Word-Art etc. without penalty up to a maximum of 30% of the stated word limit of the assignment as shown on the front cover. In no circumstances must your overall use of words in these tables, models etc. exceed this maximum 30% limit. **If a marker feels that you have used tables simply to avoid the word count, they may count these words as part of your word count.**

A sliding scale of penalties for excess length will be imposed according to the amount by which the limit has been exceeded.

1-10% excess	no penalty
11-20% excess	10% reduction in the mark
21-30% excess	20% reduction in the mark
>30% excess	the work will be capped at a pass i.e. 40%

NB. None of the above penalties will be used to change your mark that is above the pass mark, to one that is below the pass mark. Therefore, the *maximum penalty* for exceeding the word limit will be a reduction to a pass grade.

Please note that students' work is regularly checked to ensure that the stated word count matches to what has actually been submitted.

Academic Misconduct and Plagiarism

BUV follows Staffordshire University policy on academic misconduct and takes suspected cases of academic misconduct, including plagiarism very seriously. The penalties are severe and can in some cases result in a student not being allowed to continue their studies. Plagiarism can happen in any type of assessment where you are given the questions or tasks in advance.

The Staffordshire University policy on Academic misconduct can be found here: and you are strongly encouraged to familiarise yourself with this.

You will have committed academic misconduct if you commit any of the following:

- Plagiarism
- Self-Plagiarism
- Collusion
- False Declaration
- Fabrication or Falsification of Data
- Bribery or Intimidation
- Contract Cheating

- Examination Misconduct

Please see here for a detailed explanation of these terms or click the individual links above.

All work completed at BUV must be written in English directly by the submitting students. You must not, for example, write your work in Vietnamese, and then use online software to translate your completed work into English. This misrepresents the amount of effort you have put into completing your assessment. The use of online paraphrasing tools to assist with your writing is also forbidden. Any transgressions will be regarded as academic misconduct.

If Academic Misconduct is proven by the university, then typical punishments include:

- The grade for your assessment, being reduced to zero with the right to resubmit.
- The grade for the whole module or supermodule being reduced to zero with the right to resubmit.
- Failure of the entire level of which you are currently studying with a requirement to re-start the level at the next opportunity
- Failure of the award and/or termination of your studies at the University. Your ability to reapply and enrol again at the university may also be restricted, normally for two whole academic years.

BUV encourages you to refer to appropriate academic sources, as long as you reference these correctly, and do not use too much material from the original source.
You must use the Harvard referencing system for all your assignments, unless you are told otherwise by your Module Leader. If you do not know how to do this, please follow the guidelines given at this web-site address:
http://www.staffs.ac.uk/support_depts/infoservices/learning_support/refzone/harvard/

BUV employs Turnitin to assess all submissions for potential academic misconduct. This software highlights to the examiners in detail where material may have either been copied directly or paraphrased from a source with correct citation.

Appendix 1: Marking Rubric Assignment 1

Criteria	% weighting	0%-39%	40%-49%	50%-59%	60%-69%	70% - 79%	80-89%	90+
Best Practices and interface	10%	The student has not built a CLI or followed best practices such as commenting or maintaining a sensible naming scheme	The student has built a basic CLI and followed some best practices, but with significant gaps or errors	The student has built a functional CLI and followed most best practices with some minor errors	The student has built a functional CLI and followed best practices with only minor errors	The student has built a functional CLI and followed best practices thoroughly with minimal errors	The student has built a functional CLI and followed best practices thoroughly with minimal errors In every consideration of this element you have met or exceeded the lecturers expectations for the level	The student has built a functional CLI and followed best practices thoroughly with no errors You have greatly exceeded the expectations set for this element and demonstrated knowledge above and beyond what was expected
OOP	35%	The program does not use an OOP solution	The program has implemented classes but not demonstrated use of Encapsulation, inheritance, or polymorphism	The program has implemented classes and demonstrated a use of Encapsulation, inheritance, or polymorphism	The program has implemented classes and demonstrated two distinct uses of Encapsulation, inheritance, and polymorphism	The program has implemented classes and demonstrated consistent use of Encapsulation, inheritance, and polymorphism throughout	The program has implemented classes and demonstrated consistent use of Encapsulation, inheritance, and polymorphism throughout In every consideration of this element you have met or exceeded the lecturers expectations for the level	The program has implemented classes and demonstrated consistent use of Encapsulation, inheritance, and polymorphism throughout You have greatly exceeded the expectations set for this element and demonstrated knowledge above and beyond what was expected

Use cases	35%	See Appendix 2 for full breakdown	See Appendix 2 for full breakdown	See Appendix 2 for full breakdown	See Appendix 2 for full breakdown	See Appendix 2 for full breakdown	See Appendix 2 for full breakdown	See Appendix 2 for full breakdown
Demonstration	10%	Student cannot answer questions on how the code works	Student can explain their code but with minor errors	Student is capable of discussing their code and what it does but cannot provide any further detail	Student discusses their code confidently. The student can propose some alternative solutions to the problems with prompting and support	Student discusses their code confidently and without hesitation. The student can propose alternative solutions to the problems and demonstrates mastery of the project	Student discusses their code confidently and without hesitation. The student can propose alternative solutions to the problems and demonstrates mastery of the project In every consideration of this element you have met or exceeded the lecturers expectations for the level	Student discusses their code confidently and without hesitation. The student can propose alternative solutions to the problems and demonstrates mastery of the project You have greatly exceeded the expectations set for this element and demonstrated knowledge above and beyond what was expected
Adherence to University policies including generative AI policy (10%)	10%	BUV policies including the Generative AI policy, formatting and submission policies have not been followed.	Adherence to all BUV policies including the Generative AI policy, formatting and submission policies with significant mistakes that impact upon the quality of the work significantly	Adherence to all BUV policies including the Generative AI policy, formatting and submission policies with mistakes that impact upon the quality of the work	Adherence to all BUV policies including the Generative AI policy, formatting and submission policies with insignificant mistakes that do impact upon the quality of the work in a minor way	Adherence to all BUV policies including the Generative AI policy, formatting and submission policies with insignificant mistakes that barely impact upon the quality of the work	Adherence to all BUV policies including the Generative AI policy, formatting and submission policies with insignificant mistakes that do not impact upon the quality of the work	Adherence to all BUV policies including the Generative AI policy, formatting and submission policies

Appendix 2: Use Cases Marking Rubric - Assignment 1

Task	Explanation	Available Marks
Get a student grade for a specific assignment	Write a function that accepts a student ID, Module ID and Assignment ID and returns the student grade	5
Get a student grade for a module	Write a function that accepts a student ID, Module ID and returns the module grade based on the weighted average	15
Get a final degree classification	Write a function that accepts a student ID and returns the final degree classification.	15

Appendix 3: Examples

Calculating Grades for an assignment

- Assessment 4 is a report is worth 70%
- Assessment 7 is an exam is worth 30%
- We are calculating the average for student 36

Student 36 got a mark of 68 for Assessment 4 and scored 60 out of 65 for Assessment 7

The grade for Assessment 4 is 68 – Reports are marked on a scale out of 100 and are already counted as grades

The grade for Assessment 7 is 92 – Exams are marked by dividing the maximum available marks (65) by the awarded mark (60) represented by a percentage and rounding.

$$\frac{Awarded\ Marks}{Maximum\ Mark} \times 100$$

In this example

$$\frac{60}{65} \times 100 = 0.92307 \ldots \times 100 = 92\%$$

Calculating Weighted Grade for a module

- Module 2 has components
- Assessment 4 is a report is worth 70%
- Assessment 7 is an exam is worth 30%
- We are calculating the average for student 36

From the previous example

Student 36 got a mark of 68 for Assessment 4 and scored 60 out of 65 for Assessment 7

The grade for Assessment 4 is 68

The grade for Assessment 7 is 92

To calculate the weighted grade– We multiply each graded component by the weighting and add them together.

$$G_{\text{final}} = \sum_{i=1}^{n} (G_i \times W_i)$$

where:

- G_{final} is the final grade.
- G_i is the grade for the i-th component.
- W_i is the weighting of the i-th component.
- n is the total number of graded components.

In this example n is = 2 as there are two components

$$G_{\text{final}} = (G_1 \times W_1) + (G_2 \times W_2)$$

$$G_{\text{final}} = (68 \times 0.70) + (92 \times 0.30)$$

$$G_{\text{final}} = 47.6 + 27.6 = 75.2$$

The Average Weighted grade for Student A on Module 2 was 75%

Calculating Weighted Grade for a module with non-submission

- Module 5 has 2 components
- Assessment 9 is a Presentation worth 90%
- Assessment 16 is an exam is worth 10%
- We are calculating the average for student 15

Student 15 got a mark of 90 for Assessment 9 but there is no line in the database where AssessmentID == 16 and StudentID == 15. This means there is no submission for Student 15 on Assessment 16.

Since there was a non-submission the Weighted Grade for the module should be 0. There is no need to do further calculations.

Determining Final Degree Classification

- Student 25 is about to graduate.
- At Level 4 (Year 1) they completed modules – 1,3,4,6,7,9
- At Level 5 (Year 2) they completed modules – 20,21,23,24,26,27
- At Level 6 (Year 3) they completed modules – 32,33,34,35,36,37

To complete this section effectively you need to be calculating the weighted grade for modules effectively. The final degree classification only includes grades achieved at levels 5 and 6. The table below has calculated the weighted grade for the modules

Module	Level	Grade
20	5	48
21	5	58
23	5	55
24	5	52
26	5	75
27	5	79
32	6	67
33	6	63
34	6	72
35	6	79
36	6	70
37	6	41

To calculate the Final degree classification, we first remove the lowest scoring module. In this case Module 37 it is not included in any future calculation.

Next we calculate the arithmetic mean for each level. We sum the grades at each level and divide by the number of modules being summed

$$G_{\text{final}} = (M_5 \times 0.30) + (M_6 \times 0.70)$$

Where:

- M_5 is the Mean of Grades at level 5
- M_6 is the Mean of Grades at level 5

M_5 in this example = $\frac{48+58+55+52+75+79}{6}$ = 61.1666667

M_6 in this example = $\frac{67+63+72+79+70}{5}$ = 70.2

$G_{final} = (61.166667 * 0.3) + (70.2 * 0.7)$

$G_{final} = 67.49$

We can finally look up the classification table. This is a 2nd class 1st Division